



Aishu R 251001007-2025-IT ▾

A2**Started on** Friday, 2 October 2026, 7:45 PM**State** Finished**Completed on** Friday, 2 October 2026, 7:46 PM**Time taken** 46 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence: [-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int a[n], dp[n], max = 1;
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11         dp[i] = 1;
12     }
13
14     for (int i = 1; i < n; i++) {
15         for (int j = 0; j < i; j++) {
16             if (a[j] <= a[i] && dp[j] + 1 > dp[i])
17                 dp[i] = dp[j] + 1;
18         }
19
20         if (dp[i] > max)
21             max = dp[i];
22     }
23
24     printf("%d", max);
25
26     return 0;
27 }
```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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